

PSP - Aurora Global Database

- [Scope](#)
- [Context](#)
- [Lazy loading](#)
- [Aurora Global Endpoint](#)
- [Audit database](#)
- [Monolith database](#)
 - [Phase I](#)
 - [Phase II](#)
- [Aurora Global database Status and Lag check](#)
- [Production Aurora database endpoints](#)
 - [Monolith](#)
 - [Audit](#)
- [Staging Aurora database endpoints](#)
 - [Monolith](#)
 - [Audit](#)
- [Aurora Postgres global db Switchover](#)
- [Aurora Postgres global db Failover](#)
- [Aurora Global Database Monitoring](#)

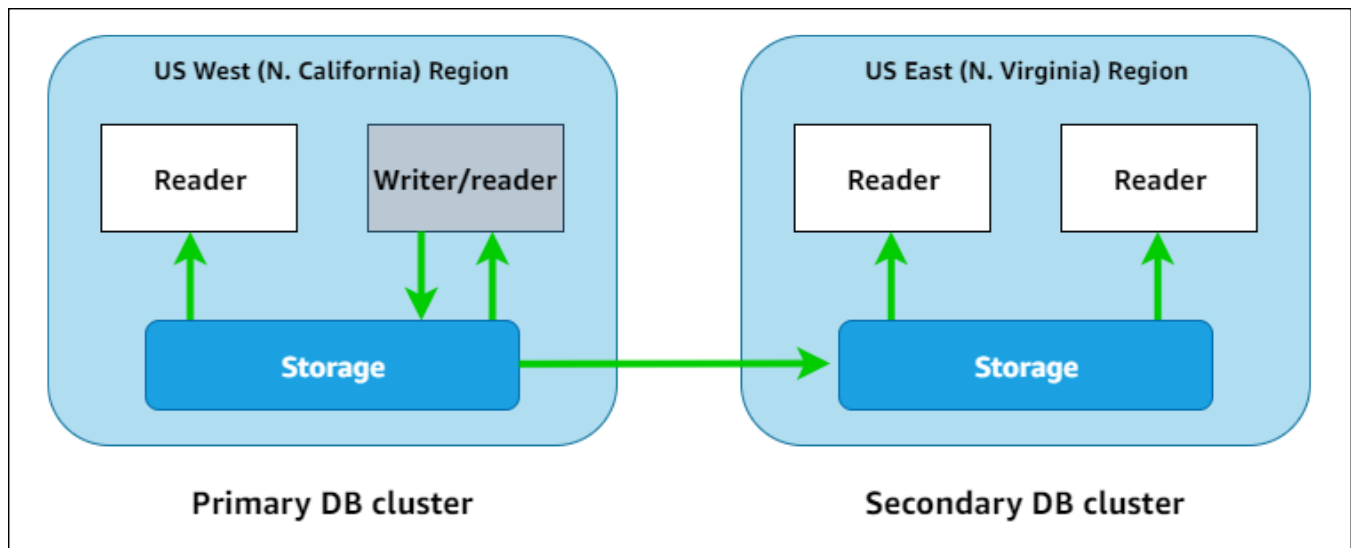
Scope

This document presents an overview of the Aurora Global Database and outlines the necessary steps for setting it up using the IPS. It includes the provisioning of the audit database via the IPS. Furthermore, it provides a strategy for transitioning a monolithic database to a path paved by the IPS.

Context

The disaster recovery for the PSP Aurora Postgres database will be facilitated through the utilization of an Aurora Global Database. Amazon's Aurora Global Database feature allows an individual Aurora database to expand across multiple AWS regions. This enhances the speed of global read operations, improves data durability, and makes the disaster recovery process much simpler. The disaster recovery plan for the PSP database will be established using the Aurora Global Database in the us-east-2 region to ensure cross-regional disaster recovery.

Aurora Postgres Global database

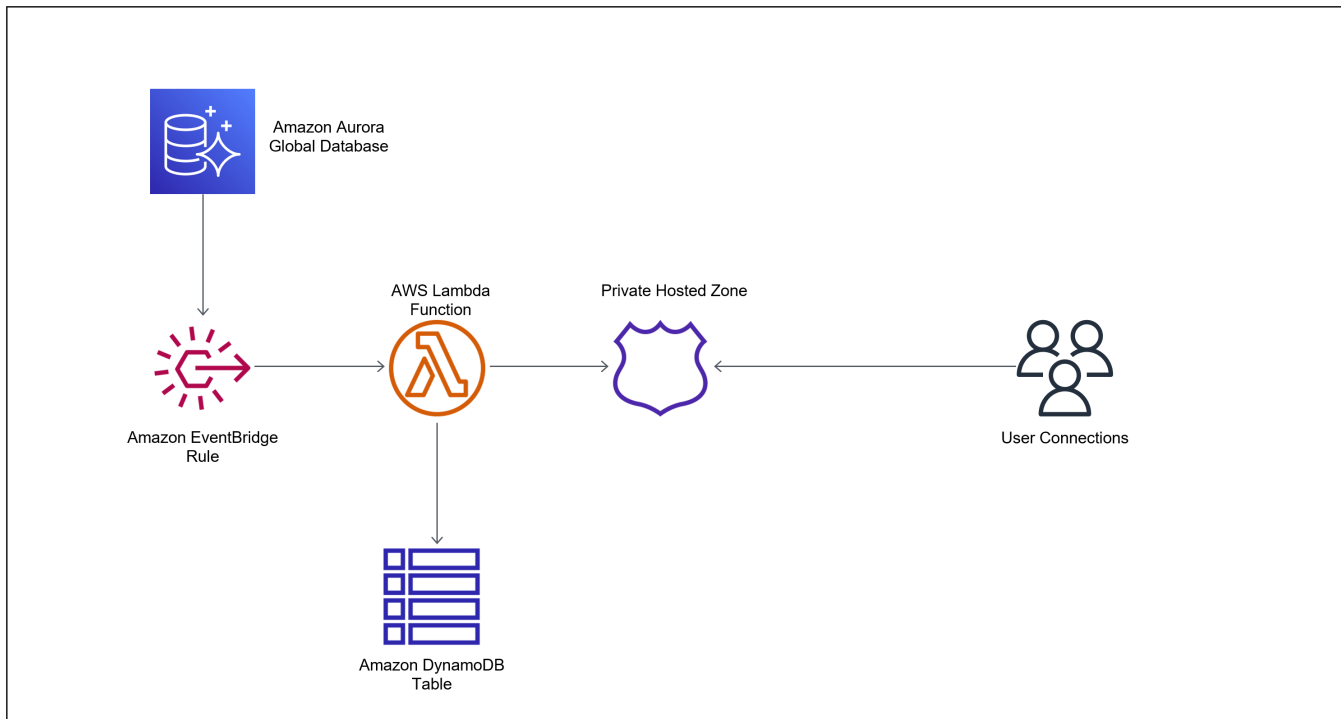


Lazy loading

Since Aurora uses AWS proprietary storage volume, the concept of Lazy Loading is NOT there in Aurora. In Aurora, once you have restored the snapshot, all the databases data files will reside in cluster volume and you will not face lazy loading issue as compared to RDS. Amazon Aurora's continuous backup and restore techniques are optimised to avoid variation in restore times. They also help the cluster's storage volume to reach full performance as soon as the cluster becomes available. This is true for Aurora Global as well, regardless of adding another region, or restoring snapshots.

Aurora Global Endpoint

In this PSP will use the Route 53 CNAME endpoint to establish a connection with the database writer. In case of a managed planned failover, below solution automates updating of CNAME, redirecting it to the writer endpoint of the primary cluster currently active in the global database. This provides an additional layer of disaster recovery and business continuity capabilities.



Audit database

Build Aurora global database in us-east-2 via IPS as below.

Monolith database

Since for Monolith database, its not build via IPS paved path, build aurora global database in 2 phase as below :

Phase I

Enable Aurora global database manually.

Phase II

Onboard Monolith database to IPS paved path.

Aurora Global database Status and Lag check

- Lag check

```

select durability_lag_in_msec/1000
from (
SELECT CASE
    WHEN '-1' = durability_lag_in_msec THEN 'Primary'
    ELSE 'Secondary'
END AS global_role,durability_lag_in_msec,rpo_lag_in_msec
FROM aurora_global_db_status()) a where global_role='Secondary';
  
```

Production Aurora database endpoints

Monolith

	Category	Applications	Database Custom Endpoint		Instance Mapping		Global Database Endpoint
			us-west-2	us-east-2	us-west-2	us-east-2	
1	Writer (OLTP /OLAP)	PSP, PAPI, Pubsub, All read-write batchjobs	psp-prod-uw02.cluster-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ue02.cluster-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-uw02	psp-prod-ue021	psp-apg-prod-02.sbg-psp-prod.a.intuit.com
2	Reader (OLTP)	Payroll DM, VMP, Equifax	psp-prod-uw02-vmp.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ue02-vmp.cluster-custom-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-uw022, psp-prod-uw023	psp-prod-ue022, psp-prod-ue023	psp-apg-prod-02-vmp.sbg-psp-prod.a.intuit.com
3	Reader (OLAP)	RJF, Read only batchjobs	psp-prod-uw02-rjf.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ue02-rjf.cluster-custom-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-uw023	psp-prod-ue023	psp-apg-prod-02-rjf.sbg-psp-prod.a.intuit.com
4	Reader (OLAP)	Finance Reporting, Downstream Services	psp-prod-uw02-rpt.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ue02-rpt.cluster-custom-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-uw024	psp-prod-ue024	psp-apg-prod-02-rpt.sbg-psp-prod.a.intuit.com
5	Reader (OLAP)	Datalake	psp-prod-uw02-datalake.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ue02-datalake.cluster-custom-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-uw024	psp-prod-ue024	Writer : psp-apg-prod-02.sbg-psp-prod.a.intuit.com Reader :psp-apg-prod-02-datalake.sbg-psp-prod.a.intuit.com

Audit

	Category	Applications	Database Custom Endpoint		Instance Mapping		Global Database Endpoint
			us-west-2	us-east-2	us-west-2	us-east-2	
1	Writer (OLTP)	PSP	psp-prod-ibob.cluster-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ibobdr.cluster-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-ibob1	psp-prod-ibob2	psp-apg-prod-ibob.sbg-psp-prod.a.intuit.com
2	Reader	PSP	psp-prod-ibob.cluster-ro-cjls0bohfgpq.us-west-2.rds.amazonaws.com	psp-prod-ibobdr.cluster-ro-cerpnqmbpq9a.us-east-2.rds.amazonaws.com	psp-prod-ibobdr1	psp-prod-ibobdr2	psp-apg-prod-ibob-ro.sbg-psp-prod.a.intuit.com

Staging Aurora database endpoints

Monolith

	Category	Applications	Database Custom Endpoint		Instance Mapping		Global Database Endpoint
			us-west-2	us-east-2	us-west-2	us-east-2	
1	Writer (OLTP /OLAP)	PSP, PAPI, Pubsub, All read-write batchjobs	ppsp-stg-pitparmo.cluster-cjls0bohfgpq.us-west-2.rds.amazonaws.com		ppsp-stg-pitparmo1	ppsp-stg-ue2-pitparmo1	psp-apg-stg-pitmo.sbg-psp-prod.a.intuit.com
2	Reader (OLTP)	Payroll DM, VMP, Equifax	ppsp-stg-vmp.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com		ppsp-stg-pitparmo2	ppsp-stg-ue2-pitparmo2	psp-apg-stg-pitmo-vmp.sbg-psp-prod.a.intuit.com
3	Reader (OLAP)	RJF, Read only batchjobs	ppsp-stg-rjf.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com		ppsp-stg-pitparmo2	ppsp-stg-ue2-pitparmo2	psp-apg-stg-pitmo-rjf.sbg-psp-prod.a.intuit.com
4	Reader (OLAP)	Finance Reporting, Downstream Services	ppsp-stg-rpt.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com		ppsp-stg-pitparmo2	ppsp-stg-ue2-pitparmo2	psp-apg-stg-pitmo-rpt.sbg-psp-prod.a.intuit.com
5	Reader (OLAP)	Datalake	ppsp-stg-datalake.cluster-custom-cjls0bohfgpq.us-west-2.rds.amazonaws.com		ppsp-stg-pitparmo2	ppsp-stg-ue2-pitparmo2	psp-apg-stg-pitmo-datalake.sbg-psp-prod.a.intuit.com

Audit

	Category	Applications	Database Custom Endpoint	Instance Mapping	Global Database Endpoint
--	----------	--------------	--------------------------	------------------	--------------------------

			us-west-2	us-east-2	us-west-2	us-east-2	
1	Writer (OLTP)	PSP	psp-par-aud.cluster-cjls0bohfgpq.us-west-2.rds.amazonaws.com		psp-par-aud1	psp-par-ue2-aud1	psp-apg-stg-aud.sbg-ppsp-prod.a.intuit.com
2	Reader	PSP	psp-par-aud.cluster-ro-cjls0bohfgpq.us-west-2.rds.amazonaws.com		psp-par-aud2	psp-par-ue2-aud2	psp-apg-stg-aud-ro.sbg-ppsp-prod.a.intuit.com

Aurora Postgres global db Switchover

before switchover

DB identifier	Status	Role	Engine	Region & AZ
psp-test-global	Available	Global database	Aurora PostgreSQL	2 regions
psp-test	Available	Primary cluster	Aurora PostgreSQL	us-west-2
psp-test-instance-1	Available	Writer instance	Aurora PostgreSQL	us-west-2c
psp-test-instance-2	Available	Reader instance	Aurora PostgreSQL	us-west-2b
psp-testdr	Available	Secondary cluster	Aurora PostgreSQL	us-east-2
psp-test-instance-dr1	Available	Reader instance	Aurora PostgreSQL	us-east-2b

Switchover time : 6 mins

Switchover Events

Time	System notes
December 11, 2023, 22:39 (UTC+05:30)	Global switchover to DB cluster psp-testdr in Region us-east-2 completed.
December 11, 2023, 22:37 (UTC+05:30)	New primary DB cluster psp-testdr in Region us-east-2 was successfully promoted.
December 11, 2023, 22:36 (UTC+05:30)	Old primary DB cluster psp-test in Region us-west-2 successfully shut down.
December 11, 2023, 22:33 (UTC+05:30)	Global switchover to DB cluster psp-testdr in Region us-east-2 started.

Time	System notes
December 11, 2023, 22:39 (UTC+05:30)	Completed failover to DB instance: psp-test-instance-dr1
December 11, 2023, 22:39 (UTC+05:30)	Global switchover to DB cluster psp-testdr in Region us-east-2 completed.
December 11, 2023, 22:38 (UTC+05:30)	Promoting previous primary again: psp-test-instance-dr1
December 11, 2023, 22:37 (UTC+05:30)	New primary DB cluster psp-testdr in Region us-east-2 was successfully promoted.
December 11, 2023, 22:36 (UTC+05:30)	Old primary DB cluster psp-test in Region us-west-2 successfully shut down.
December 11, 2023, 22:33 (UTC+05:30)	Global switchover to DB cluster psp-testdr in Region us-east-2 started.

after switchover

DB identifier	Status	Role	Engine	Region & AZ
psp-test-global	Available	Global database	Aurora PostgreSQL	2 regions
psp-test	Available	Secondary cluster	Aurora PostgreSQL	us-west-2
psp-test-instance-1	Available	Reader instance	Aurora PostgreSQL	us-west-2c
psp-test-instance-2	Available	Reader instance	Aurora PostgreSQL	us-west-2b
psp-testdr	Available	Primary cluster	Aurora PostgreSQL	us-east-2
psp-test-instance-dr1	Available	Writer instance	Aurora PostgreSQL	us-east-2b

Aurora Postgres global db Failover

Aurora Global Database Monitoring

- Add monitoring for Aurora db cluster added in us-east-2